

NextGenC — Gridded SOC maps of Finland

Procedural note for the kriged SOC map products that accompany the NextGenC report. Companion to `build_soc_matrices.R` (point matrices) and `build_soc_maps.R` (this product).

1 What this produces

For each of the 6 models (SP1, TP2, TP3, Yasso07, Yasso15, Yasso20), the posterior-predictive SOC at the ~ 447 calibration plots is interpolated to a regular grid covering Finland, **one layer per year, 1985–2024 (40 years)**. Two multiband GeoTIFFs per model, plus a multi-model ensemble pair (**14 files total**):

File	Bands	Content
<code><MODEL>.SOC_mean.tif</code>	40	Kriged mean total SOC (tC/ha), band $n = \text{year } 1984 + n$
<code><MODEL>.SOC_sd.tif</code>	40	Total uncertainty SD (tC/ha), same grid/year as mean band
<code>ENSEMBLE.SOC_mean.tif</code>	40	Six-model ensemble mean SOC (tC/ha)
<code>ENSEMBLE.SOC_sd.tif</code>	40	Ensemble total SD (within + between-model + spatial)

Band names carry the year (e.g. `y1985`), written into GeoTIFF metadata so labels travel with the file. Band n of every SD stack is the uncertainty for band n of the matching mean stack (same year, grid, and land mask).

2 Grid and storage

- **CRS:** ETRS89-LAEA Europe (EPSG:3035), metres; equal-area. The grid is snapped to the EEA reference grid (cell edges on multiples of the 2 km resolution from the LAEA origin). Kriging is performed natively in this projection; source inputs (plot coordinates, boundary, peat/water masks) are reprojected from ETRS-TM35FIN (EPSG:3067).
- **Resolution:** 2 km. Grid 330×589 (~ 194 k cells, of which $\sim 44\%$ land).
- **Mask:** clipped to the Finland boundary from `maakunta.sf.rds`.
- **Format:** GeoTIFF, DEFLATE-compressed, float32.
- **Footprint:** ~ 0.12 GB for all 560 maps (14 files).

The 2 km choice is a deliberate compromise: the ~ 447 plots have a **mean spacing of ~ 27.5 km**, so 2 km already oversamples the information content — finer grids render smoother but add no real spatial detail. Disk budget (1–2 GB) is not the binding constraint; data density is.

3 Method

Kriging is done **in log space** and back-transformed, because SOC is positive and right-skewed and HIKET’s error model is multiplicative — this keeps the map uncertainty multiplicative and consistent with the calibration. Per model $\times$ year:

1. **Inputs.** Plot mean and SD come from the posterior-predictive bundles (the same `soc_mean / soc_sd` columns used by `build_soc_matrices.R`); plot coordinates (`x.ETRS`, `y.ETRS`) come from `Data/model_inputs/site`, joined on `plot_id`.
2. **Krige the mean.** Ordinary kriging of `log(soc_mean)` over the grid $\rightarrow$ interpolated mean surface plus the **kriging prediction variance** v_{krige} (small near plots, large in data-sparse areas, e.g. Lapland).
3. **Krige the model uncertainty.** Ordinary kriging of `log(soc_sd)` $\rightarrow$ a surface of the model’s own posterior-predictive uncertainty.
4. **Combine (option C).** Total uncertainty fuses both sources,

$$\text{sd}_{\text{total}} = \sqrt{v_{\text{krige}} + \text{kriged_modelSD}^2}$$

(combined in log space, then back-transformed). This propagates **model uncertainty** and **spatial-coverage uncertainty** into a single SD layer — neither source is silently dropped.

5. **Back-transform & mask.** Exponentiate to tC/ha, clip to the land mask, write the two stacks.

Variogram: **one pooled model per model**, reused for all 40 years (the spatial structure of SOC is essentially static year-to-year; this is stable and avoids per-year auto-fit failures). `set.seed(2025)` as elsewhere in HIKET.

Peat & water exclusion. The maps target **mineral-soil forest SOC**. Peat/water calibration plots (`site_raw` soil codes Tu/Ve) are dropped from the kriging *input* so they do not pull the interpolated surface, and grid cells are masked to NoData where they are **>50 % peatland** or **>50 % water**. Peat: Luke MS-NFI site-main-class raster (`paatyyppe`, 16 m, EPSG:3067; spruce/pine mire + treeless peatland), aggregated to the 2km grid as a peat fraction of *the whole cell* (water in the denominator, so lake-dominated cells do not read as peat). This site-type definition captures drained peatland forest (*turvekangas*), which land-cover products miss (~10 % of land masked). Water: SYKE Ranta10 lake polygons (the sea is outside the land polygon). See `Data/Peat_extension/` and `Data/Water_extension/`. In the thumbnail figures peat renders **black** and water/sea **white**.

4 Ensemble (six-model) product

The ensemble is formed *on the grid*, after each model has been kriged — i.e. the per-model mean and total-SD stacks above are combined cell-by-cell, equal-weighted ($w_i = 1/6$). With $M = 6$ model mean surfaces m_i and total-SD surfaces s_i (each s_i already folds in that model’s kriging variance and posterior uncertainty, §3 step 4):

$$m_{\text{ens}} = \frac{1}{M} \sum_i m_i, \quad s_{\text{ens}} = \sqrt{\underbrace{\frac{1}{M} \sum_i s_i^2}_{\text{within-model}} + \underbrace{\frac{1}{M-1} \sum_i (m_i - m_{\text{ens}})^2}_{\text{between-model (structural)}}}.$$

This is the law of total variance: the ensemble deviation fuses **(i)** each model’s own predictive uncertainty (which already carries **(iii)** the spatial/kriging uncertainty) with **(ii)** the *structural spread* between models — the among-model disagreement that is the central HIKET question. The between-model term is the scientifically interesting one: where it dominates, the map is uncertain because the models *disagree*, not because the data are sparse. Combination is done in the same (log/back-transformed) space as the per-model SDs for consistency.

Note: the between-model term is also informative on its own; if a structural-disagreement map is wanted later, export $\sqrt{\frac{1}{M-1} \sum_i (m_i - m_{\text{ens}})^2}$ as a separate stack — one extra line in the script.

5 How to run

from the project root

Rscript Reporting/NextgenC_report/build_soc_maps.R

Reads the same bundles `<- c(...)` `RUN_ID` map as `build_soc_matrices.R`. Writes the 12 Geo-TIFFs into `Reporting/NextgenC_report/`.

Dependencies: `terra`, `gstat`, `sf` (and `sp`). Posterior-predictive bundles must be present in `Calibration_real_data_transient/runs/`, and `Data/model_inputs/{site_raw.csv, maakunta.sf.rds}` must exist.

6 Maintenance notes

- **TP3 RUN_ID is provisional.** As in `build_soc_matrices.R`, the TP3 bundle is the pre-fix (Euler) predictive. Swap the TP3 RUN_ID to the exact-integrator re-calibration when it is

synced from Puhti, then re-run.

- **Adding years / plots.** No code change — the script reads year columns and plot rows from the bundles directly.
- **Changing resolution.** Edit the single `RES_M <- 2000` constant. Storage scales as $1/\text{res}^2$: 1 km $\approx$ 0.4 GB, 5 km $\approx$ 0.02 GB (all 560 maps, compressed).
- **Reading the output.** `terra::rast("SP1_SOC_mean.tif")` gives a 40-layer `SpatRaster`; layer names are the years. Same for the `_sd` file.